CANADIAN MATH KANGAROO CONTEST

Grade 3 and 4 Questions and Solutions
Year 2025

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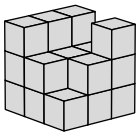
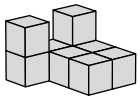
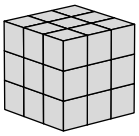
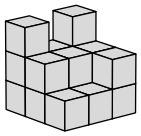
September 13, 2025

MATH
KANGAROO

CANADIAN MATH KANGAROO CONTEST PROBLEMS

PART A: EACH CORRECT ANSWER IS WORTH 3 POINTS

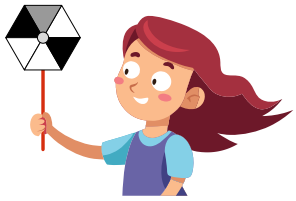
1. Mia is joining small cubes, adding one at a time, to build a $3 \times 3 \times 3$ cube. She took pictures at 5 different moments. What does Mia's fourth picture look like?

(A)  **(B)**  **(C)**  **(D)**  **(E)** 

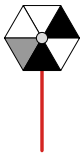
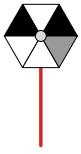
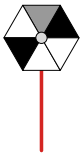
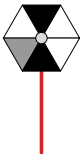
2. Simona writes the four digits 2, 0, 2, 5 in the four boxes: $\square + \square - \square + \square$.
Which order would give her the largest result?

(A) 0, 2, 2, 5 **(B)** 0, 5, 2, 2 **(C)** 2, 5, 2, 0 **(D)** 5, 0, 2, 2 **(E)** 5, 2, 0, 2

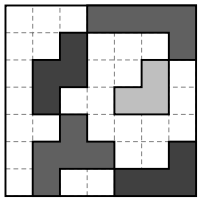
3. Larissa spins her pinwheel:



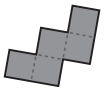
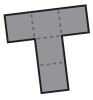
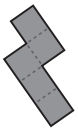
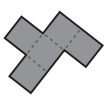
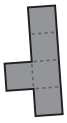
Which of the pinwheels below is hers?

(A)  **(B)**  **(C)**  **(D)**  **(E)** 

4. Andy placed some pieces on the board.

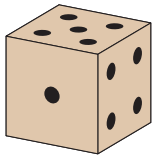
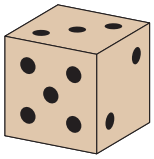
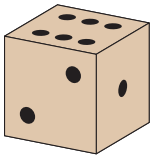
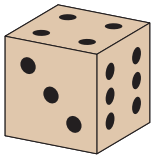
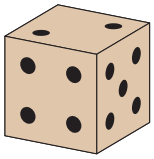


Which of the following pieces **cannot** fit into the empty white squares?"

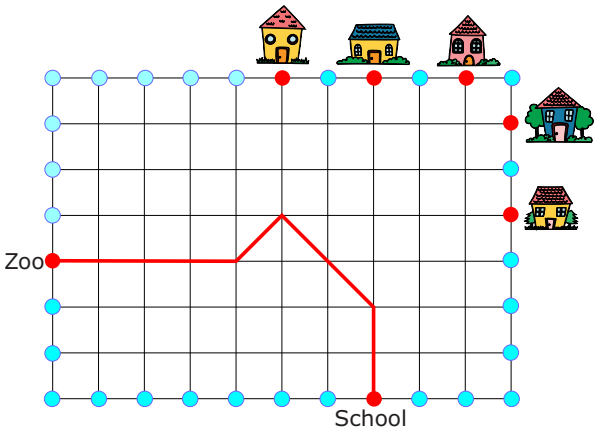
(A)  **(B)**  **(C)**  **(D)**  **(E)** 

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5. On a standard die, the total number of spots on two opposite faces is always 7.
Which of the dice shown could be a standard die?

(A)  (B)  (C)  (D)  (E) 

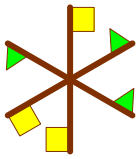
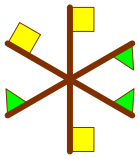
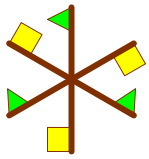
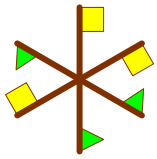
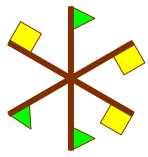
6. Kenny the Kangaroo jumps from the School to the Zoo as follows: ↑ 2, ↖ 2, ↙ 1, ← 4, as shown.



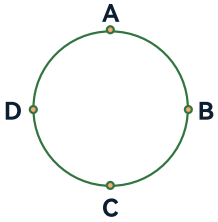
Then, he jumps from the Zoo as follows: → 3, ↓ 1, ↗ 4, ↑ 1.
Which house will he get to?

(A)  (B)  (C)  (D)  (E) 

7. Which shape can Jorge build with these 3 rods? 

(A)  (B)  (C)  (D)  (E) 

8. In a circular playground, Zhenzhen and Lele start at point A at the same time. Zhenzhen walks clockwise (↻), while Lele runs counterclockwise (↺). They first meet at point B, then again at point C, a third time at point D, and finally return to point A for their fourth meeting.
How many laps did Lele run in total?

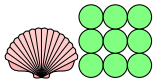
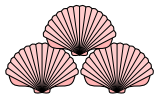
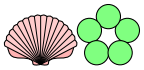
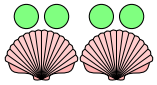


(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

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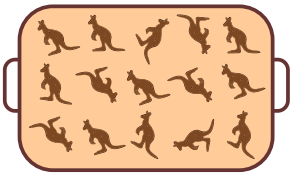
PART B: EACH CORRECT ANSWER IS WORTH 4 POINTS

9. Nico and his little sister pay with shells and marbles in their playshop. Each shell has a value of 6 and each marble has a value of 1. Which of the following has a total value of 16?

(A)  (B)  (C)  (D)  (E) 

10. Anna, Bonnie and Caspar have some kangaroo cookies on their plates, as shown.

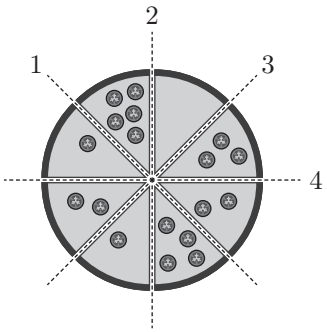




They then share the remaining 15 cookies on the tray so that everyone now has the same number of cookies on their plates.
How many more cookies does Bonnie get?

(A) 4 (B) 5 (C) 6 (D) 7 (E) 8

11. Tom wishes to slice a pizza into two halves.
He also wishes to have the same number of tomatoes on each half.
It is possible for him to do this with two different cuts.
Along which lines could he cut?

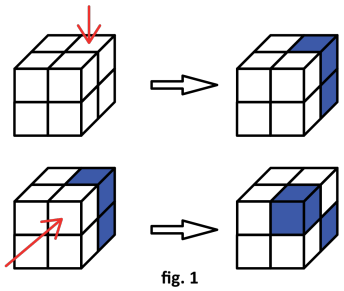


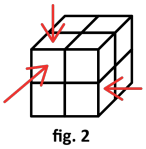
(A) 1 or 3 (B) 1 or 4

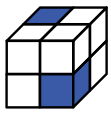
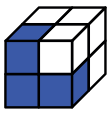
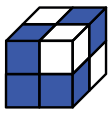
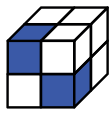
(C) 2 or 3 (D) 2 or 4

(E) 3 or 4

12. A large cube is made up of eight small cubes. Touching the face of a small cube changes the color of that cube and the cube on its opposite face: white cubes turn blue, and blue cubes turn white.
In fig. 1, you can see what happens when the cube is touched at the two indicated places.
What will the original white cube look like after being touched three times as shown in fig. 2 (the order of touches doesn't matter)?

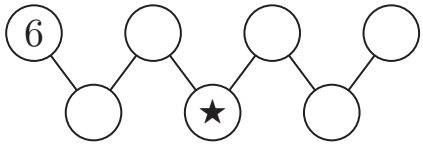

fig. 1


fig. 2

(A)  (B)  (C)  (D)  (E) 

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13. Maria fills the circles with the numbers 1, 2, 3, 4, 5, 6, and 7. Each bottom circle contains the sum of the two numbers in the circles directly above it.
What number is in the circle marked with ★?



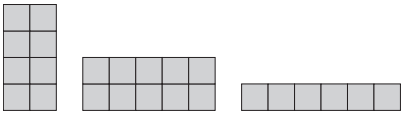
(**A**) 2 (**B**) 3 (**C**) 4 (**D**) 5 (**E**) 7

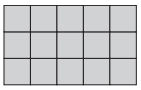
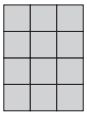
14. In the morning, five friends had identical fully-charged mobile phones. By the evening, Bob had used his phone as much as Ann and Cristina together. Bob's battery was empty. David had not used his phone at all.
Which phone belonged to Edward?



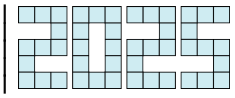
(**A**) 1 (**B**) 2 (**C**) 3 (**D**) 4 (**E**) 5

15. Bob makes a square from four rectangular pieces. Three of the pieces he uses are shown.
Which of the following is the fourth piece he uses?



(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

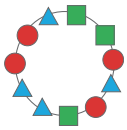
16. The number 2025 in the figure is reflected over the line to its left.
Which of the following is the reflected number?

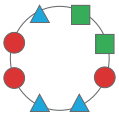
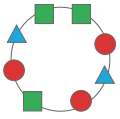
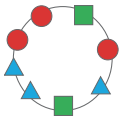
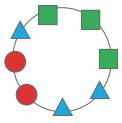
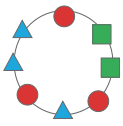


(**A**) 2025 (**B**) 5202 (**C**) 5205 (**D**) 2505 (**E**) 5052

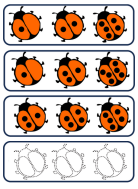
PART C: EACH CORRECT ANSWER IS WORTH 5 POINTS

17. Amy removes a group of three connected shapes from the figure, including a triangle, a square, and a circle. She then reconnects the remaining shapes in the correct order. Which of the following options will she get?



(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

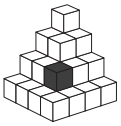
18. Six ladybugs have 1, 2, 3, 4, 5 or 6 spots each. Marta took four photos of them in groups of three. Each ladybug appeared the same number of times in the photos. Three of the photos, along with the outline of the fourth photo, are shown.
How many spots do the three ladybugs in Marta's fourth photo have in total?

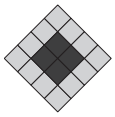


(**A**) 9 (**B**) 10 (**C**) 11 (**D**) 12 (**E**) 23

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19. Leonia has built a pyramid using black and grey cubes, arranging them so that no face of one cube touches a face of another cube with the same color. One of the black cubes is shown in the figure.
What will Leonia's pyramid look like from above?



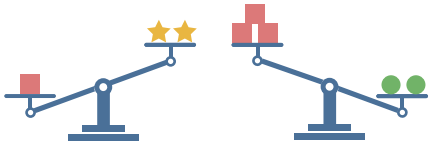
(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

20. The figure shows a calendar page for one month, with the dates missing. The sum of the dates for the two shaded days is 29.
On which day of the week does the first day of the month fall?

Mon	Tue	Wed	Thu	Fri	Sat	Sun

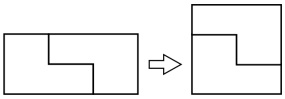
(**A**) Monday (**B**) Tuesday (**C**) Wednesday (**D**) Thursday (**E**) Sunday

21. Two scales are used to weigh three different objects, and the result is shown. Each type of object has a unique weight, with possible values of 1, 2, 3, 4, or 5.
What is the weight of one ■?



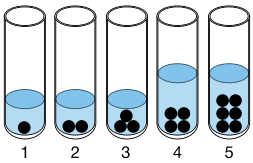
(**A**) 1 (**B**) 2 (**C**) 3 (**D**) 4 (**E**) 5

22. The figure shows a rectangle divided into two identical parts by a zigzag line, and these parts can form a square.
If the length of the rectangle is 27 centimeters, what is the width of the rectangle in centimeters?



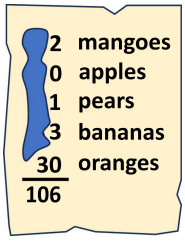
(**A**) 10 (**B**) 12 (**C**) 15 (**D**) 16 (**E**) 18

23. Identical balls are placed in 5 identical tubes, as shown. Water is then added to each tube. The water levels in tubes 1, 2, and 3 are the same, while the water levels in tubes 4 and 5 are also the same, but twice as high as those in the first three tubes.
After all the balls are removed, which tube has the least amount of water?



(**A**) Tube 1 (**B**) Tube 2 (**C**) Tube 3 (**D**) Tube 4 (**E**) Tube 5

24. Lily has recorded the number of fruits of each variety she owns. Unfortunately, some digits are covered by paint. In total, she has 106 fruits. Two varieties have the same number of fruits, and one variety has twice as many fruits as another. Additionally, she has more than 10 fruits of each variety.
How many bananas does she have?



(**A**) 3 (**B**) 13 (**C**) 23 (**D**) 33 (**E**) 43

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CMKC 2025 Grade 3-4 Answers

PART A					
1	**A**	B	C	D	E
2	A	B	C	D	**E**
3	A	B	C	D	**E**
4	A	**B**	C	D	E
5	**A**	B	C	D	E
6	A	**B**	C	D	E
7	A	B	C	D	**E**
8	A	B	**C**	D	E

PART B					
9	A	B	C	D	**E**
10	A	**B**	C	D	E
11	A	B	C	**D**	E
12	**A**	B	C	D	E
13	A	B	**C**	D	E
14	A	**B**	C	D	E
15	A	B	C	D	**E**
16	A	B	C	**D**	E

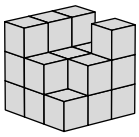
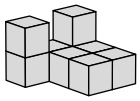
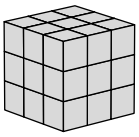
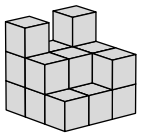
PART C					
17	**A**	B	C	D	E
18	A	B	C	**D**	E
19	A	B	C	**D**	E
20	A	B	C	**D**	E
21	A	B	**C**	D	E
22	A	**B**	C	D	E
23	A	B	**C**	D	E
24	A	**B**	C	D	E

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CANADIAN MATH KANGAROO CONTEST SOLUTIONS

PART A: EACH CORRECT ANSWER IS WORTH 3 POINTS

1. Mia is joining small cubes, adding one at a time, to build a $3 \times 3 \times 3$ cube. She took pictures at 5 different moments. What does Mia's fourth picture look like?

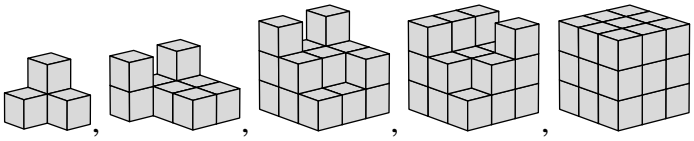
(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

Solution

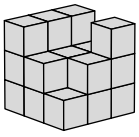
Mia builds the big cube by joining small cubes, adding one at a time.

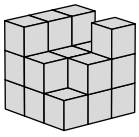
The numbers of small cubes increases.

Hence, her photos show this order:



or: (D), (B), (E), (A), (C) in this order.

Her fourth picture must be  .

Answer: (A) 

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2. Simona writes the four digits 2, 0, 2, 5 in the four boxes: $\square + \square - \square + \square$.
Which order would give her the largest result?

(**A**) 0, 2, 2, 5 (**B**) 0, 5, 2, 2 (**C**) 2, 5, 2, 0 (**D**) 5, 0, 2, 2 (**E**) 5, 2, 0, 2

Solution

Solution 1:

One of the digits is subtracted, and the other three digits are added.

So, the largest result would be when the third digit that is subtracted is the smallest, i.e. 0.

Only option (E) 5, 2, 0, 2 has a third digit 0.

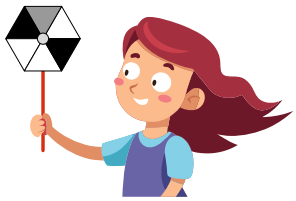
$$\boxed{5} + \boxed{2} - \boxed{0} + \boxed{2}.$$

Solution 2:

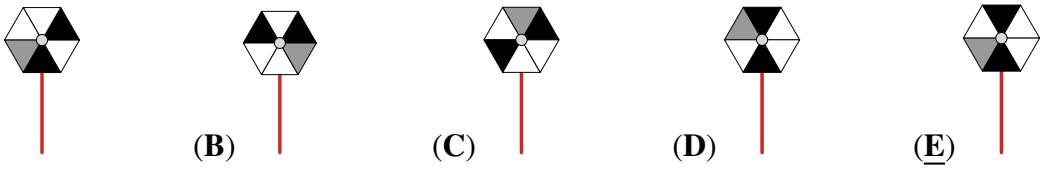
It is also possible to compute all five results to find the largest:
$0 + 2 - 2 + 5 = 5$;
$0 + 5 - 2 + 2 = 5$;
$2 + 5 - 2 + 0 = 5$;
$5 + 0 - 2 + 2 = 5$;
$5 + 2 - 0 + 2 = 9$.

Answer: (E) 5, 2, 0, 2

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3. Larissa spins her pinwheel:
Which of the pinwheels below is hers?



(A) **(B)** **(C)** **(D)** **(E)**

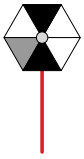
Solution

In Larissa's pinwheel, two black fields are positioned opposite each other.

This is seen only in (C), (D), and (E).

Additionally, the gray field in Larissa's pinwheel is positioned clockwise next to a black field.

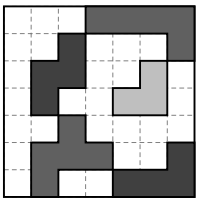
This is only present in (E), making it the correct answer.



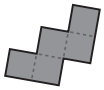
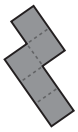
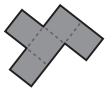
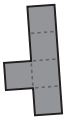
Answer: (E)

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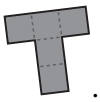
4. Andy placed some pieces on the board.

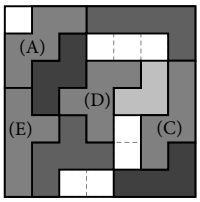


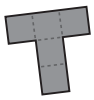
Which of the following pieces **cannot** fit into the empty white squares?"

(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

Solution

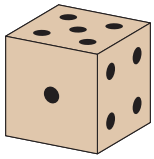
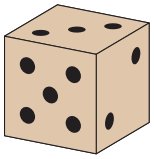
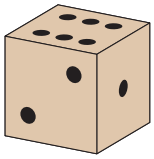
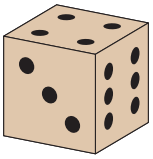
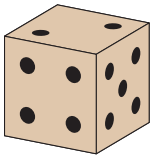
All other pieces fit, even at the same time, but there is no space for the shape (B)  .



Answer: (B) 

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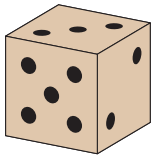
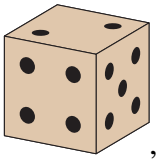
5. On a standard die, the total number of spots on two opposite faces is always 7.
Which of the dice shown could be a standard die?

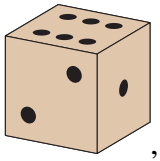
(A)  (B)  (C)  (D)  (E) 

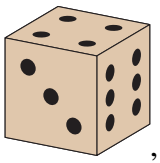
Solution

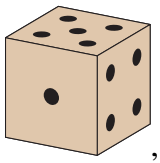
On standard dice the total number of spots on two opposite faces is always 7.

This means that:
1 and 6 are on opposite faces; 2 and 5 are on opposite faces; and 3 and 4 are on opposite faces.

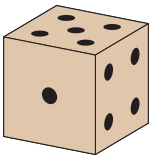
In (B)  and (E)  , 2 and 5 spots are on neighbouring faces and hence these are not standard dice.

In (C)  , 1 and 6 spots are on neighbouring faces and hence this is not a standard die.

In (D)  , 3 and 4 are on neighbouring faces, so this is not a standard die.

In (A)  , we see only one of the numbers 1 and 6, 2 and 5, and 3 and 4, so for each of these pairs the other number can be on the opposite face.

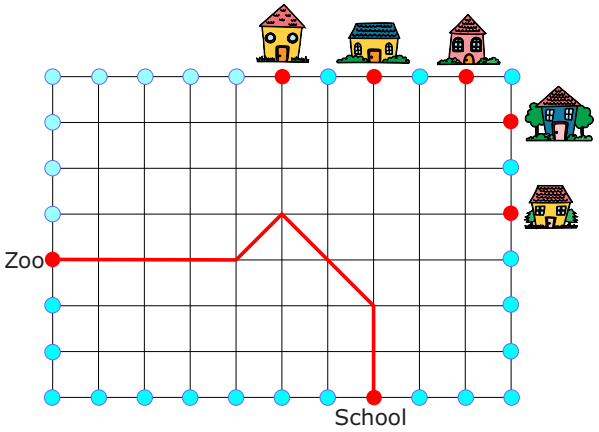
Only (A) can be a standard die.

Answer: (A) 

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6. Kenny the Kangaroo jumps from the School to the Zoo as follows: ↑ 2, ↖ 2, ↙ 1, ← 4, as shown.

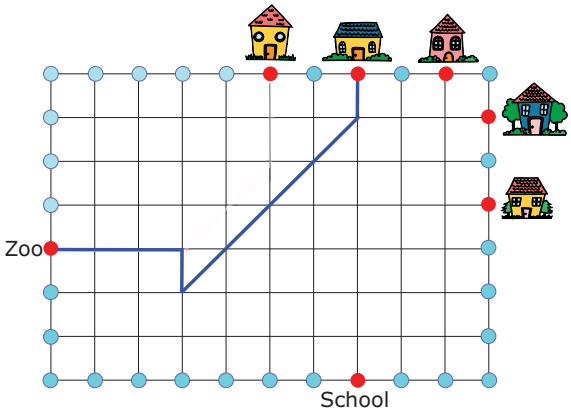


Then, he jumps from the Zoo as follows: → 3, ↓ 1, ↗ 4, ↑ 1.
Which house will he get to?

(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

Solution

Starting at the Zoo, we draw the path that Kenny the Kangaroo covers according to the instructions: → 3, ↓ 1, ↗ 4, ↑ 1 (see below).



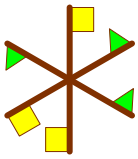
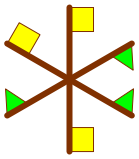
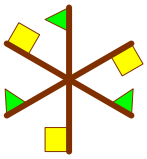
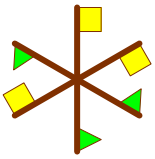
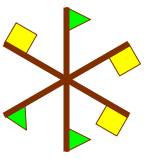
Kenny the Kangaroo gets to house  .

Answer: (B) 

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7. Which shape can Jorge build with these 3 rods?



(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

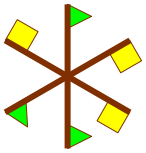
Solution

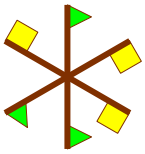
The rod with the two green triangles has both triangles on the same side.

In option (C), there is no rod with two triangles.

In options (A), (B), and (D), the triangles are placed on opposite sides of the rod.

Therefore, Jorge cannot build shapes (A), (B), (C), and (D) with these 3 rods.

He can only build shape (E)  with these 3 rods.

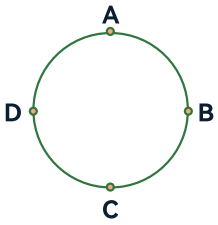
Answer: 

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8. In a circular playground, Zhenzhen and Lele start at point A at the same time. Zhenzhen walks clockwise (↻), while Lele runs counterclockwise (↺). They first meet at point B, then again at point C, a third time at point D, and finally return to point A for their fourth meeting.

How many laps did Lele run in total?



(**A**) 1 (**B**) 2 (**C**) 3 (**D**) 4 (**E**) 5

Solution

Solution 1:

Lele ran counterclockwise (↺) from A to B, then from B to C, from C to D, and finally from D back to A, where he met Zhenzhen, who had completed her first lap walking clockwise (↻).

By the time they both returned to point A for their fourth meeting, Zhenzhen had walked one (1) full lap, while Lele had run three (3) laps.

Solution 2:

Lele and Zhenzhen together cover one full lap before meeting.

Zhenzhen walked one full lap while meeting Lele four times.

In total, both Zhenzhen and Lele covered four laps – one lap by Zhenzhen and three laps by Lele.

Solution 3:

Lele runs 3/4 of the track, and Zhenzhen runs 1/4 before they meet.

Since they meet back at point A, Lele must have run a whole number of laps. This first happens after 4 times, because 4 times 3/4 laps gives 3 laps.

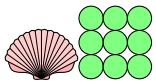
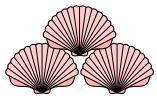
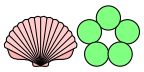
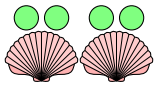
So, Lele completes 3 full laps when they meet at point A for the first time.

Answer: (C) 3

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PART B: EACH CORRECT ANSWER IS WORTH 4 POINTS

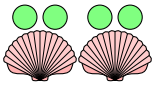
9. Nico and his little sister pay with shells and marbles in their playshop. Each shell has a value of 6 and each marble has a value of 1. Which of the following has a total value of 16?

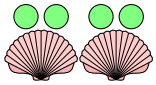
(A)  (B)  (C)  (D)  (E) 

Solution

We calculate the value for each collection.

option	shells	marbles	total value
(A)	2 shells $\times 6 = 12$	1 marble $\times 1 = 1$	$12 + 1 = 13$
(B)	1 shell $\times 6 = 6$	9 marbles $\times 1 = 9$	$6 + 9 = 15$
(C)	3 shells $\times 6 = 18$	0 marbles $\times 1 = 0$	$18 + 0 = 18$
(D)	1 shell $\times 6 = 6$	5 marbles $\times 1 = 5$	$6 + 5 = 11$
(E)	2 shells $\times 6 = 12$	4 marbles $\times 1 = 4$	$12 + 4 = \mathbf{16}$

(E)  has a total value of 16.

Answer: (E) 

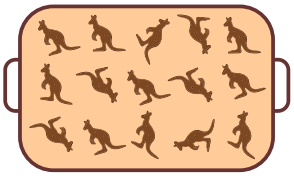
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10. Anna, Bonnie and Caspar have some kangaroo cookies on their plates, as shown.



They then share the remaining 15 cookies on the tray so that everyone now has the same number of cookies on their plates.

How many more cookies does Bonnie get?



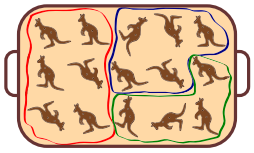
(**A**) 4 (**B**) 5 (**C**) 6 (**D**) 7 (**E**) 8

Solution

Solution 1:

Anna, Bonnie, and Caspar have 3, 4, and 5 cookies, respectively. If Anna gets 2 more cookies and Bonnie gets 1 more, they will each have 5 cookies. There are 12 cookies left to be equally divided, so each child receives 4 more cookies.

Than means Bonnie must get $1 + 4 = 5$ more cookies in total.



Solution 2:

Another approach is to first calculate the total number of cookies each child should receive. They already have 3, 4, and 5 cookies, and with the 15 additional cookies, there are a total of $3 + 4 + 5 + 15 = 27$ cookies. Each child should receive $27 \div 3 = 9$ cookies.

Thus, Bonnie must receive $9 - 4 = 5$ more cookies.

Solution 3:

If Caspar gives a cookie to Anna, all three would have the same number of cookies. This means Bonnie is not part of the exchange - she already has the right amount.

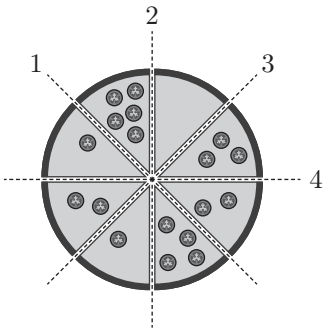
So, the remaining 15 cookies can be shared equally, with Bonnie getting 5, and the rest balanced between Anna and Caspar.

Bonnie gets 5 more cookies.

Answer: (B) 5

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11. Tom wishes to slice a pizza into two halves.
He also wishes to have the same number of tomatoes on each half.
It is possible for him to do this with two different cuts.
Along which lines could he cut?



(**A**) 1 or 3 (**B**) 1 or 4

(**C**) 2 or 3 (**D**) 2 or 4

(**E**) 3 or 4

Solution

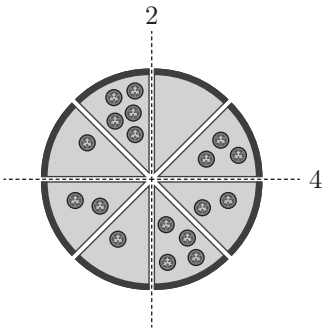
Solution 1:

Cut 1 gives $5 + 3 + 2 = 10$ tomatoes above it and $4 + 1 + 2 + 1 = 8$ tomatoes below it.

Cut 2 gives $5 + 1 + 2 + 1 = 9$ and $0 + 3 + 2 + 4 = 9$ tomatoes on each half.

Cut 3 gives $5 + 1 + 2 = 8$ and $3 + 2 + 4 + 1 = 10$ tomatoes on each half.

Cut 4 results in $3 + 5 + 1 = 9$ and $2 + 4 + 1 + 2 = 9$ tomatoes on each half.



Solution 2:

Note that knowing that there are 18 tomatoes in total allows us to check that there are 9 tomatoes on one half only.

Answer: (D) 2 or 4

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12. A large cube is made up of eight small cubes. Touching the face of a small cube changes the color of that cube and the cube on its opposite face: white cubes turn blue, and blue cubes turn white. In fig. 1, you can see what happens when the cube is touched at the two indicated places.

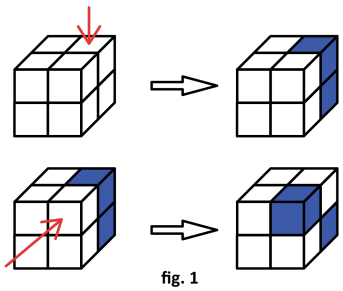


fig. 1

What will the original white cube look like after being touched three times as shown in fig. 2 (the order of touches doesn't matter)?

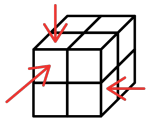
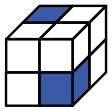
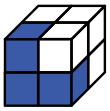
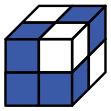
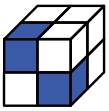
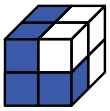


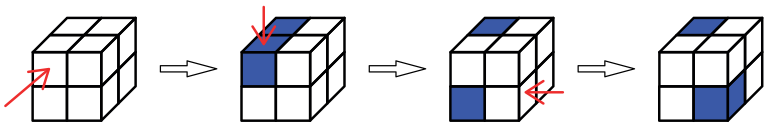
fig. 2

(A)  (B)  (C)  (D)  (E) 

Solution

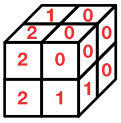
Solution 1:

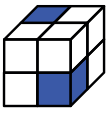
We can follow how the colors of the cubes change after each touch:



Solution 2:

Another way to see the result is to count the number of changes for each specific cube. An even number of changes will leave the color white, and an odd number will change it to blue.

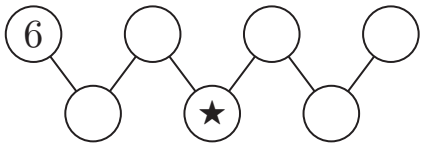


Answer: (A) 

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13. Maria fills the circles with the numbers 1, 2, 3, 4, 5, 6, and 7. Each bottom circle contains the sum of the two numbers in the circles directly above it.

What number is in the circle marked with ★?

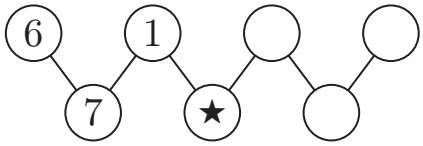


(**A**) 2 (**B**) 3 (**C**) 4 (**D**) 5 (**E**) 7

Solution

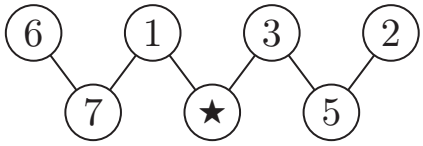
The numbers to fill are 1, 2, 3, 4, 5, and 7, with the number 6 already placed.

The number below 6 must be greater than 6, meaning the number 7 must be placed in the bottom-left circle. Since only 1 added to 6 equals 7, the number 1 must be placed to the right of 6.



The remaining numbers to be placed are 2, 3, 4, and 5. Since the two largest numbers should be placed in the bottom row, the number 4 must come from the sum of 1 + 3, as it cannot be obtained by adding any pair from 2, 3, and 5.

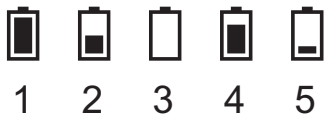
This places the number 4 in the bottom-middle position, which is the correct answer (the star), and the number 3 goes in the top row. The number 5 is then placed in the bottom-right, and the number 2 in the top-right.



Answer: (C) 4

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14. In the morning, five friends had identical fully-charged mobile phones. By the evening, Bob had used his phone as much as Ann and Cristina together. Bob's battery was empty. David had not used his phone at all. Which phone belonged to Edward?



(**A**) 1 (**B**) 2 (**C**) 3 (**D**) 4 (**E**) 5

Solution

Since Bob ran out of power, his phone must be phone 3.

David had not used his phone at all, so his phone is phone 1.

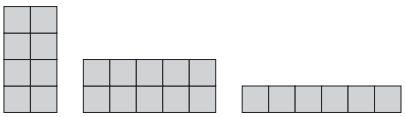
Only phones 4 and 5 have battery usage that adds up to a full charge, meaning they must belong to Ann and Cristina.

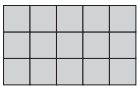
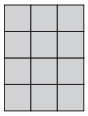
This leaves phone 2, which must belongs to Edward.

Answer: (B) 2

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15. Bob makes a square from four rectangular pieces.
Three of the pieces he uses are shown.
Which of the following is the fourth piece he uses?



(A)  (B)  (C)  (D)  (E) 

Solution

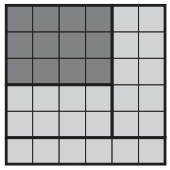
The largest piece has a side length of 6 squares, so the large square must be at least $6 \times 6 = 36$ small squares.

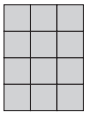
The three given pieces total $8 + 10 + 6 = 24$ small squares. The fourth piece must therefore consist of at least $36 - 24 = 12$ squares (or 25 squares if the large square's side length is 7).

Since none of the options contain 25 or more squares, the large square must be 36 squares with a side length of 6. Only the pieces in (D) and (E) are made of 12 small squares.

To find the missing piece, we must construct the large 6×6 square. We have to turn the first and second pieces. The missing rectangle is 4 squares wide and 3 squares high.

Thus, the fourth piece is option (E). The large square Bob can make is given below.

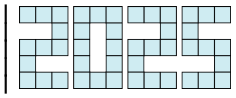


Answer: (E) 

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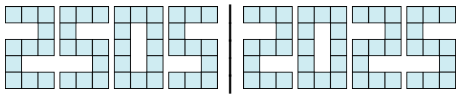
16. The number 2025 in the figure is reflected over the line to its left.
Which of the following is the reflected number?



(**A**) 2025 (**B**) 5202 (**C**) 5205 (**D**) 2505 (**E**) 5052

Solution

The reflected number over the line to its left is 2505, as shown in the figure below.

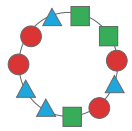


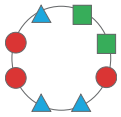
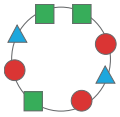
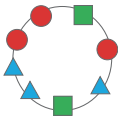
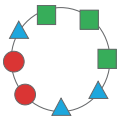
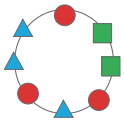
Answer: (D) 2505

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PART C: EACH CORRECT ANSWER IS WORTH 5 POINTS

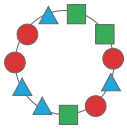
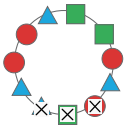
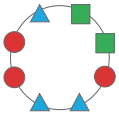
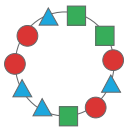
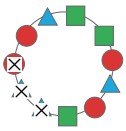
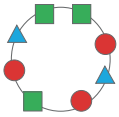
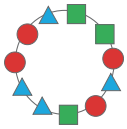
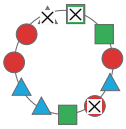
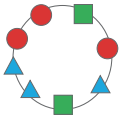
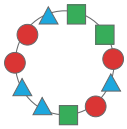
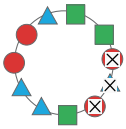
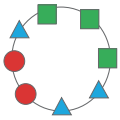
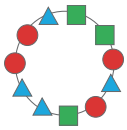
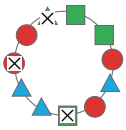
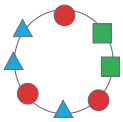
17. Amy removes a group of three connected shapes from the figure, including a triangle, a square, and a circle. She then reconnects the remaining shapes in the correct order. Which of the following options will she get?



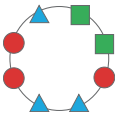
(A)  (B)  (C)  (D)  (E) 

Solution

Amy can obtain the five shapes by removing three pieces as shown below.

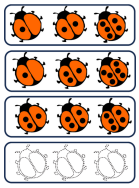
Option	Original	Removal	Result
(A)			
(B)			
(C)			
(D)			
(E)			

Only (A) is a result of the removal of a group of three connected shapes from the figure, including a triangle, a square, and a circle.

Answer: (A) 

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18. Six ladybugs have 1, 2, 3, 4, 5 or 6 spots each. Marta took four photos of them in groups of three. Each ladybug appeared the same number of times in the photos. Three of the photos, along with the outline of the fourth photo, are shown.



How many spots do the three ladybugs in Marta's fourth photo have in total?

(**A**) 9 (**B**) 10 (**C**) 11 (**D**) 12 (**E**) 23

Solution

Solution 1:

The ladybugs with 1, 3, and 5 spots appear twice in the first three photos, while those with 2, 4, 6 appear only once.

Therefore, the 4th photo must have ladybugs with 2, 4 and 6 spots, for a total of $2+4+6=12$ spots.

Solution 2:

The six ladybugs have a total of

$$1+2+3+4+5+6=(1+6)+(2+5)+(3+4)=7+7+7=21$$

spots.

Since the ladybugs appear equally in the four photos in groups of three, each must appear twice, giving a total of 42 spots across all photos.

Next, we calculate the spots in the first three photos:

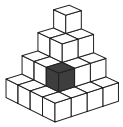
$$(1+2+5)+(1+3+6)+(3+4+5)=8+10+12=30.$$

Therefore, the total number of spots for the three ladybugs in Marta's fourth photo is $42-30=12$.

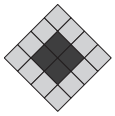
Answer: (D) 12

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19. Leonia has built a pyramid using black and grey cubes, arranging them so that no face of one cube touches a face of another cube with the same color. One of the black cubes is shown in the figure.



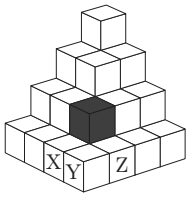
What will Leonia's pyramid look like from above?

(A)  **(B)**  **(C)**  **(D)**  **(E)** 

Solution

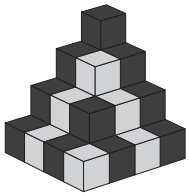
Solution 1:

The cube beneath the black cube must be grey. This means that cubes X and Z, which are adjacent to this grey cube in the bottom layer, must both be black. As a result, cube Y must be grey. Among the five options, only option (D) reflects this configuration. Examining the neighboring cubes further confirms that option (D) is correct.



Solution 2:

As a first step, we color the blocks so that no two blocks of the same color share a faces. There is only one possibility to do this:



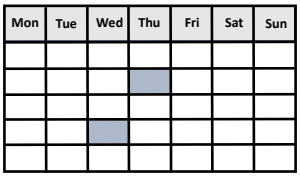
Viewing the structure from above, we can see that the correct solution is option (D).

Answer: (D) 

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20. The figure shows a calendar page for one month, with the dates missing. The sum of the dates for the two shaded days is 29.

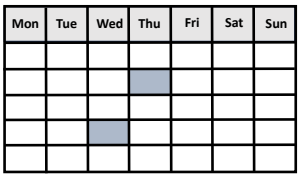
On which day of the week does the first day of the month fall?



(**A**) Monday (**B**) Tuesday (**C**) Wednesday (**D**) Thursday (**E**) Sunday

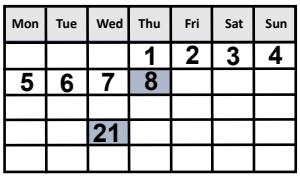
Solution

Counting the days from the first shaded square to the second, we see that the second date is 13 days after the first.



We need two numbers that differ by 13 and sum to 29. These numbers can only be 21 and 8 because their difference is $21 - 8 = 13$ and their sum is $21 + 8 = 29$.

Since the first day of the month is exactly one week before the 8th, the days of the week must align, making the first day Thursday.

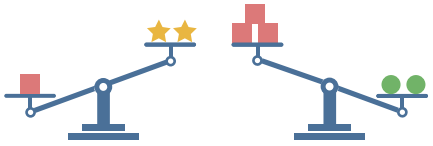


Answer: (D) Thursday

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21. Two scales are used to weigh three different objects, and the result is shown. Each type of object has a unique weight, with possible values of 1, 2, 3, 4, or 5.

What is the weight of one ■?



(**A**) 1 (**B**) 2 (**C**) 3 (**D**) 4 (**E**) 5

Solution

The smallest possible mass for a star is 1.

The first weighing shows that one square is heavier than two stars, so the square must weigh more than 2, specifically 3, 4, or 5.

The second weighing shows that two circles are heavier than three squares, so the three squares must weigh at least 9. This means the two circles must weigh more than 9, and since each object weighs 1, 2, 3, 4, or 5, one circle must weigh 5.

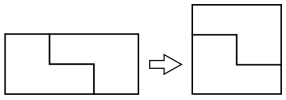
One square must weigh either 3 or 4. If a square weighs 4, then three squares would be heavier than two circles (since $3 \times 4 > 2 \times 5$), but this contradicts the first weighing.

Therefore, the weight of one square (■) must be 3.

Answer: (C) 3

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22. The figure shows a rectangle divided into two identical parts by a zigzag line, and these parts can form a square.

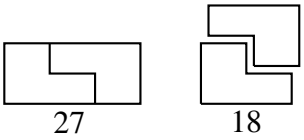


If the length of the rectangle is 27 centimeters, what is the width of the rectangle in centimeters?

(**A**) 10 (**B**) 12 (**C**) 15 (**D**) 16 (**E**) 18

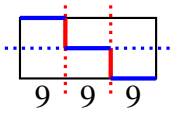
Solution

The rectangle is divided into two equal parts by the zigzag line, as shown below.

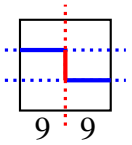


The vertical dashed lines divide the rectangle into three equal smaller rectangles, each with a length that is one-third of the total length of the larger rectangle.

Similarly, the horizontal dashed line divides the rectangle into two equal smaller rectangles, each with a width that is half the total width of the larger rectangle.



This ensures that the pieces can be arranged to form a square with a side length of 18.



$18 = 9 + 9 = 6 + 6 + 6$

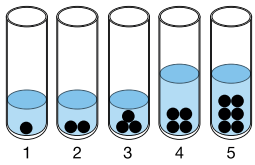
The width of the rectangle is two-thirds of 18 centimeters, which equals $12 = 6 + 6$ centimeters.

Answer: (B) 12

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23. Identical balls are placed in 5 identical tubes, as shown. Water is then added to each tube. The water levels in tubes 1, 2, and 3 are the same, while the water levels in tubes 4 and 5 are also the same, but twice as high as those in the first three tubes.

After all the balls are removed, which tube has the least amount of water?



(**A**) Tube 1 (**B**) Tube 2 (**C**) Tube 3 (**D**) Tube 4 (**E**) Tube 5

Solution

The amount of water in tube 3 is less than in tubes 1 and 2 because it contains more balls, yet the water level remains the same.

Similarly, the amount of water in tube 5 is less than in tube 4.

Now, comparing tubes 3 and 5: if we pour the contents of two tubes 3 into one tube, we match the content of tube 5.

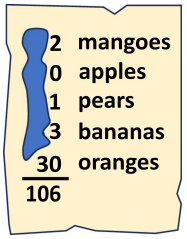
This means tube 5 contains not only twice as many balls as tube 3 but also twice as much water.

Therefore, tube 3 holds the least amount of water.

Answer: (C) Tube 3

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24. Lily has recorded the number of fruits of each variety she owns. Unfortunately, some digits are covered by paint. In total, she has 106 fruits. Two varieties have the same number of fruits, and one variety has twice as many fruits as another. Additionally, she has more than 10 fruits of each variety. How many bananas does she have?



(**A**) 3 (**B**) 13 (**C**) 23 (**D**) 33 (**E**) 43

Solution

The fruits that are equal in number, based on the last digit, are the apples and oranges, with 30 each. Based on the last digits, we can also conclude that the varieties of fruit with twice as many as another are the mangoes (*2) and the pears (*1).

Since each type of fruit has more than 10 pieces, the number of pears must be 11, 21, 31, 41, and so on, while the number of mangoes must be 22, 42, 62, and so on.

The total number of mangoes, apples, pears, and oranges is at least $22 + 30 + 11 + 30 = 93$.

Therefore, the number of bananas is at most $106 - 93 = 13$.

Since each type of fruit has more than 10 pieces, there must be exactly 13 bananas.

Answer: (B) 13

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